



Probability Distributions

DAY 2 – MERIDIAN RETAIL

Bernoulli · Binomial · Poisson · Normal – moving from **describing data** that already happened to **predicting** what is likely to happen next.

Probability Distributions

DAY 2 · MERIDIAN RETAIL

Bernoulli

One event. Two outcomes. The building block.

Binomial

Many trials. Count the successes.

Poisson

Events in a fixed window of time.

Normal

Measurements that cluster around a centre.

Same company as yesterday: Meridian Retail, same order book, four new questions. Today we move from describing what happened to predicting what comes next.

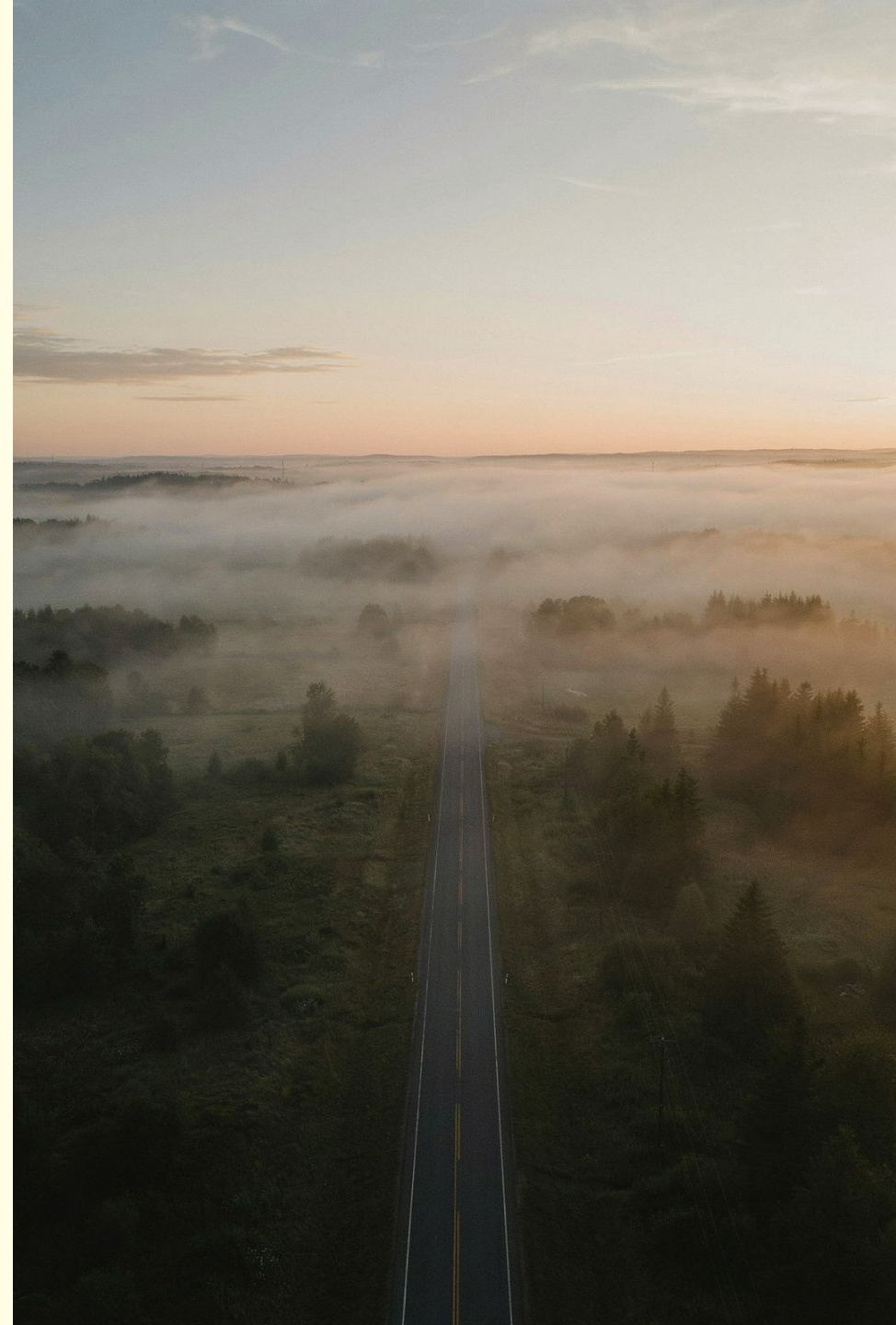
Where We Were, and Where We Are Going

Yesterday — Describing Data

Mean, median, standard deviation, skewness, kurtosis all answer the same question: **what does this data look like?** Every one of them needs the data in hand first.

Today — Predicting the Future

How many of the next 20 visitors will buy something? How many complaint calls should we staff for tomorrow morning? What fraction of parcels will be too heavy for the standard shipping rate? None of these can be answered by looking at a column. They need a **probability distribution**.



What a Probability Distribution Actually Is

A probability distribution is a rule that assigns a probability to every possible outcome. That is the whole definition. It takes two parts:

1. A List of Outcomes

For a coin: heads or tails. For complaint calls in an hour: 0, 1, 2, 3, and so on. Every possible result must be named.

2. A Probability for Each

Each outcome gets a number between 0 and 1. These probabilities **must add up to exactly 1** – the fastest way to check whether an answer is nonsense.

Once you have that rule, you can answer questions about the future without waiting for the future to arrive.

Discrete or Continuous – The First Question to Ask

This single question narrows your choice from four distributions to two before anything else.

Discrete – You Are Counting

Answers are whole numbers. Nothing in between is possible.

- Number of customers who bought: 0, 1, 2, 3
- Number of complaint calls: 0, 1, 2
- There is no such thing as 2.7 customers

Bernoulli, Binomial, Poisson are all discrete.

Continuous – You Are Measuring

Any value in a range is possible – including decimals.

- Parcel weight: 1.85 kg, 1.8513 kg
- Delivery time, temperature, order value
- Precision is limited only by your instrument

Normal is continuous.

The Four Distributions at a Glance

Everything in the rest of this deck expands one of these four rows. Keep this table in mind as a navigation anchor throughout.

Distribution	Use it when you are asking...	Type
Bernoulli	Will this one thing happen? Yes or no.	Discrete
Binomial	Out of n tries, how many will succeed?	Discrete
Poisson	How many events in a fixed window of time?	Discrete
Normal	What values will a measurement take?	Continuous

PART 1

Bernoulli

One event. Two outcomes.



Bernoulli: The Single Yes or No

The simplest distribution there is. One trial, two possible outcomes. We label one outcome **success** (written as 1) and the other **failure** (written as 0). "Success" is just a label — a loan default can be the success if that is what we are counting.



E-commerce

One customer visits the site. They either buy or they do not.



Logistics

One parcel is shipped. It either arrives on time or it does not.



Credit

One loan is issued. It either defaults or it does not.

One parameter: p , the probability of success. That is all you need to describe the entire distribution.

Bernoulli: The Formula

The Rules

$$P(\text{success}) = p$$

$$P(\text{failure}) = 1 - p$$

$$\text{Mean} = p$$

$$\text{Variance} = p(1 - p)$$

Meridian Example — Return Rate $p = 0.146$

For any single order picked at random from Meridian's order book:

- $P(\text{returned}) = \mathbf{0.146}$
- $P(\text{not returned}) = \mathbf{0.854}$
- $\text{Mean} = \mathbf{0.146}$
- $\text{Variance} = 0.146 \times 0.854 = \mathbf{0.125}$

Note: no single order is ever "0.146 returned". The mean here is a long-run rate, not a description of one order.

Where Bernoulli Is Applied

Anywhere a single event has exactly two outcomes. Every binary column in a dataset – Yes/No, Pass/Fail, 1/0 – is Bernoulli data.

Marketing

Did this visitor convert?

Credit

Did this borrower default?

Quality Control

Did this unit pass inspection?

Operations

Did this delivery meet the SLA?

HR

Did this candidate accept the offer?

Bernoulli: Advantages and Disadvantages

✓ Advantages

- The simplest possible model – one parameter, nothing to tune
- p is easy to estimate: count successes, divide by total
- It is the building block for the next three distributions

▪ Disadvantages

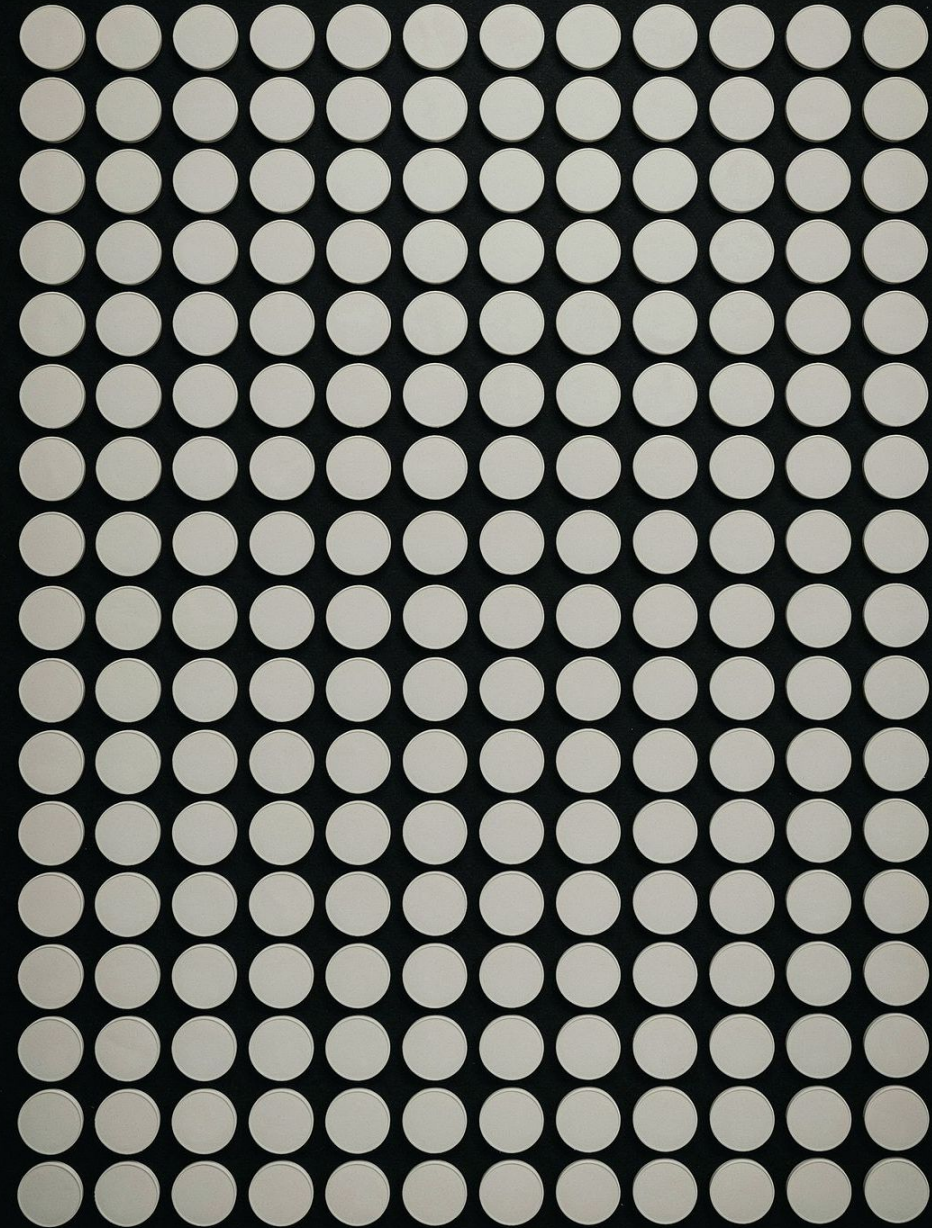
- Describes **one** trial only – almost no business question is about a single trial
- Says nothing about how many successes you get across many trials
- Forces any richer situation into a binary yes/no

📌 Knowing one visitor converts at probability 0.28 does not tell you how many of tomorrow's 500 visitors will convert. For that you need the **Binomial**.

PART 2

Binomial

Many trials. Count the successes.



Binomial: Counting Successes Across n Trials

Run the same yes/no trial n times and count how many succeed. A binomial is simply n **Bernoulli trials added together**.

20 Visitors

Arrive at the site. How many place an order?

100 Parcels

Ship from the warehouse. How many arrive late?

50 Loans

Are issued this month. How many default?

Parameter n

The number of trials – decided in advance.

Parameter p

The probability of success on each individual trial.

The Four Conditions That Must Hold

Check these before using the binomial. If any one fails, the answer will be wrong.

1 Fixed number of trials

You decide n in advance: 20 visitors, not "however many turn up".

2 Two outcomes per trial

Success or failure, nothing else.

3 Same probability every time

p does not change from trial 1 to trial 20. It breaks when a discount goes live mid-day.

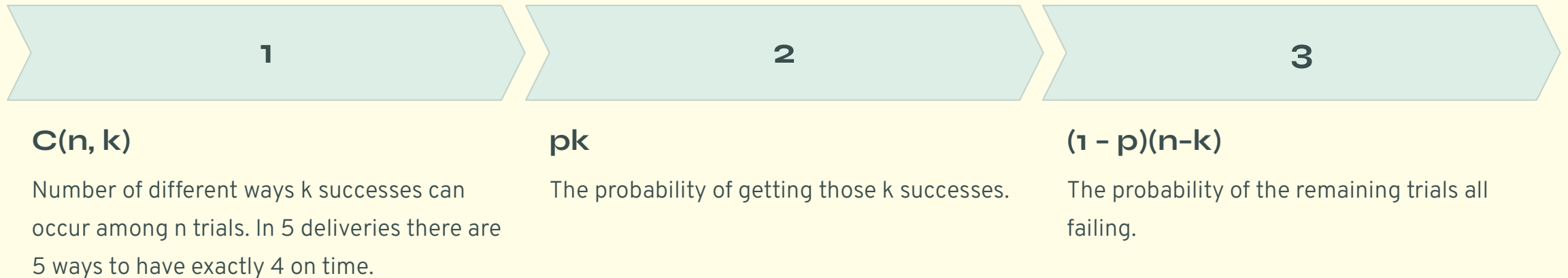
4 Trials are independent

One outcome does not affect the next. This is the condition that breaks most often – group bookings, queues, and viral products all violate it.

Binomial: The Formula

$$P(X = k) = C(n, k) \times p^k \times (1 - p)^{(n-k)}$$

Read it in three pieces, left to right:



Multiply the three together and you have the probability of exactly k successes.

Binomial: Worked Calculation

Meridian ships **5 parcels**. Each has a **90% chance** of arriving on time. What is the probability that **exactly 4** arrive on time? Here $n = 5$, $p = 0.9$, $k = 4$.

01

Count the arrangements

$C(5, 4) = 5$ – the late parcel could be any one of the five.

03

Probability the 5th is late

$0.1^1 = 0.1$

02

Probability of 4 on time

$0.9^4 = 0.6561$

04

Multiply all three

$5 \times 0.6561 \times 0.1 = 0.328$

There is a **32.8% chance** that exactly 4 of the 5 parcels arrive on time.

Binomial: Mean and Variance

5.6

Expected Conversions

Mean = 20×0.28 – the centre of the distribution.

4.03

Variance

$20 \times 0.28 \times 0.72$ – spread around that centre.

2.01

Std. Deviation

$\sqrt{4.03}$ – the typical day-to-day swing in conversions.

For 20 visitors at a 28% conversion rate, a typical day sees about 5 or 6 conversions, and the day-to-day swing is roughly 2 in either direction. Both numbers matter – the average tells you where you are, the standard deviation tells you how much to expect it to vary.

Formulas: Mean = $n \times p$ | Variance = $n \times p \times (1 - p)$ | SD = $\sqrt{(n \times p \times (1 - p))}$

Where Binomial Is Applied



Marketing and E-commerce

Conversion counts from a fixed number of visitors. Email campaigns: how many of 10,000 sends will be opened. **A/B testing – the entire framework rests on the binomial.**



Quality Control

Out of a batch of 500 units, how many will be defective. Acceptance sampling: inspect 50, reject the batch if more than 3 fail.



Finance and Credit

Out of 200 loans in a portfolio, how many will default. Number of trading days in a month that end in a loss.



Binomial: Advantages

Only Two Inputs

n and p – both easy to obtain from historical data with minimal effort.

Answers the Real Business Question

Not "will this one customer buy" but "how many of them will" – which is the number that drives decisions.

Full Picture, Not Just an Average

You can ask for the probability of any specific count, or a range of counts, not just the expected value.

Foundation of A/B Testing

Well understood, universally implemented, and immediately useful for evaluating campaigns and experiments.

Binomial: Disadvantages

Requires a Fixed n

If you do not know how many trials there will be, you cannot use it. "How many complaints tomorrow?" has no n – that is a Poisson question.

Assumes Constant p

Real conversion rates move with time of day, season, and active campaigns. A discount going live mid-day breaks this assumption immediately.

Assumes Independence

Group behaviour, queues, and word of mouth all break this – and they are common in retail.

Dangerous Failure Mode

When assumptions fail, the answer looks precise and *is wrong*. Underestimated variability is worse than no estimate at all.



PART 3

Poisson

Events in a window of time.

Poisson: Counting Events in a Fixed Window

Use this when events happen at some average rate and you want to know how many will occur in a given period. **The key difference from binomial: there is no fixed number of trials.** There is no "n". Calls simply arrive.

Support Desk

Complaint calls arrive at about 6 per hour.
How many will arrive in the next hour?

Manufacturing

A machine produces about 2 defects per shift.
What is the chance of 5 or more?

Web Traffic

A website receives about 300 visits per minute at peak.
How many in the next 30 seconds?

One parameter: λ (lambda) – the average number of events per window. That is all you need.

Poisson: The Conditions

1 Events occur independently

One call arriving does not make the next call more or less likely to arrive.

2 The average rate is constant

Over the window you are studying. **This is the one to watch.** Complaint calls are not constant across a day – there is a morning peak and a quiet afternoon. Fix: model each period separately with its own λ .

3 Two events cannot occur at exactly the same instant

Events are discrete – they can be counted and separated in time.

4 The window is fixed

Per hour, per day, per square metre, per shift. Clearly defined before you calculate.

Poisson: The Formula

$$P(X = k) = (e^{-\lambda} \times \lambda^k) / k!$$

λ (lambda)

The average number of events per window – the only input.

k

The specific number of events you are asking about.

$e = 2.71828$

A fixed mathematical constant – not something you choose.

$k!$ (factorial)

$4! = 4 \times 3 \times 2 \times 1 = 24$. Grows quickly as k increases.

You will rarely compute this by hand. What matters is knowing *when* it applies, and that λ is the only input you need to supply.

Poisson: Worked Calculation

A support desk receives an average of **3 calls per hour**. What is the probability of exactly **2 calls** in the next hour? Here $\lambda = 3$, $k = 2$.

01

Compute $e^{-\lambda}$

$$e^{-3} = \mathbf{0.0498}$$

03

Compute $k!$

$$2! = \mathbf{2}$$

02

Compute λ^k

$$3^2 = \mathbf{9}$$

04

Put it together

$$(0.0498 \times 9) / 2 = \mathbf{0.224}$$

There is a **22.4% chance** of exactly 2 calls in the next hour.

Poisson: The Property That Makes It Distinctive

Mean = λ

The average number of events is λ .

Variance = λ

The spread is also λ . **The mean and variance are the same number.**

No other common distribution does this.

A Free Diagnostic Test

Take your real data, compute the mean and the variance:

- If they are **roughly equal** → Poisson is a reasonable model
- If **variance much larger than mean** → data is overdispersed and Poisson will understate your risk

Overdispersion is common in practice – one system outage generates fifty calls at once, clustering events together and inflating variance.

Where Poisson Is Applied

The common thread: you are counting arrivals in a window, and there is no fixed number of trials.



Operations and Support

Call centre volume and staffing levels, machine breakdowns per shift, stockouts per week.



Technology

Server requests per second, errors logged per hour, packets arriving at a network switch.



Insurance and Risk

Number of claims filed per month, accident counts per region and per period.



Retail and Healthcare

Customer arrivals per 15-minute slot, which directly drives checkout staffing decisions.

Poisson: Advantages

Only One Input

λ – just the historical average. Nothing else required.

Handles the "No Fixed n" Case

The scenario that binomial cannot touch. Most real arrival problems have this shape.

Directly Answers Staffing Questions

How many agents, counters, or beds do we need to cover 95% of hours? Poisson gives you the number.

Free Model Check Built In

The mean = variance property gives you an immediate diagnostic before you commit to the model.

Works for Space as Well as Time

Defects per square metre of fabric, potholes per kilometre of road – any fixed window works.

Poisson: Disadvantages

Assumes a Constant Rate

Real demand has peaks. One daily λ applied to a morning rush will badly understate the true volume.

Events Cluster in Reality

One outage causes a burst of calls. One accident causes several claims. Independence fails silently.

Overdispersion Is Common

When variance exceeds the mean, Poisson understates the tail – exactly the part you were planning for.

Cannot Handle Seasonal Rates

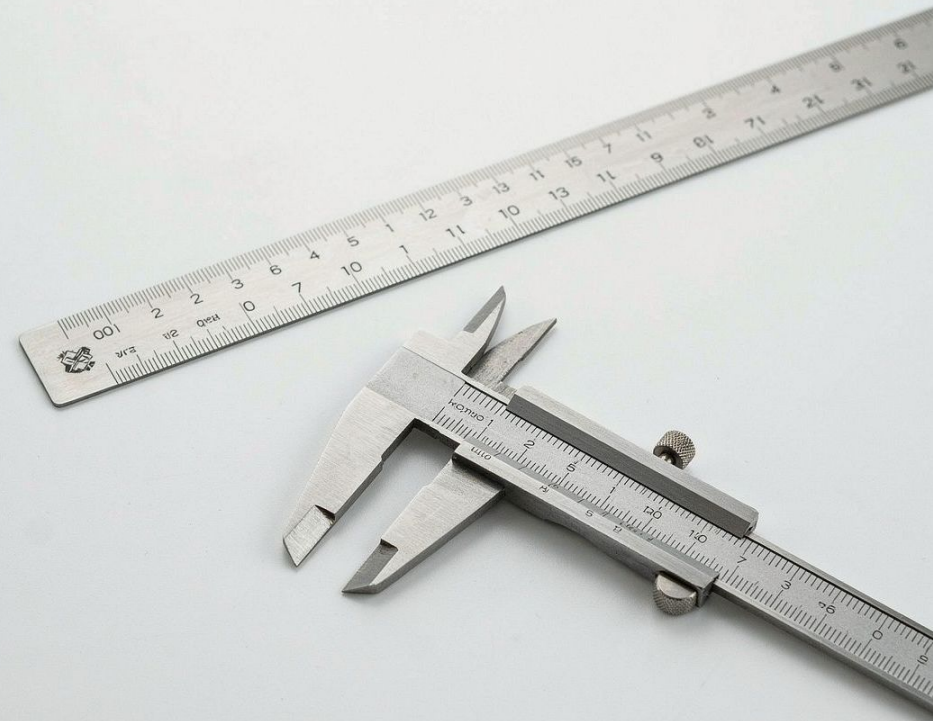
Without splitting data into separate windows. One λ across Black Friday and a quiet Tuesday is meaningless.

 When overdispersed, use the **negative binomial distribution** instead. Worth knowing the name; the mechanics are beyond today's scope.

PART 4

Normal

Measurements, not counts.



Normal: The Continuous Case

Use the normal distribution when you are **measuring** rather than counting, and values cluster around a central value with fewer observations further out.

Parameter μ (mu)

The mean – sets where the centre of the distribution sits. Change μ and the curve slides left or right.

Parameter σ (sigma)

The standard deviation – sets how wide the curve is. Change σ and it gets wider or narrower. These are the only two things that can change.

Common Applications

- Parcel weights
- Heights, temperatures, manufacturing dimensions
- Measurement errors of any kind
- Averages of large samples – even when the original data is not normal at all

Normal: What the Shape Tells You

1

Perfectly Symmetric

Mean, median, and mode all sit at the same point. Skewness is exactly 0. A long tail on one side rules it out.

2

Fully Described by Two Numbers

Give someone μ and σ and they can reconstruct the entire distribution. No other information is needed.

3

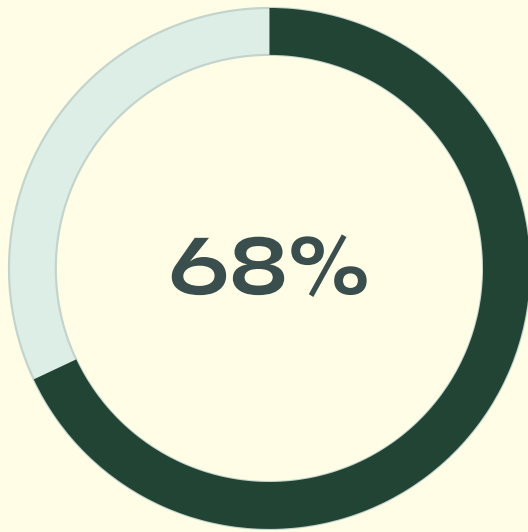
Tails Never Reach Zero

Values far from the mean are possible and very unlikely. The curve approaches but never touches the axis.

Because it is symmetric, a normal distribution can never model something with a long tail on one side – which rules out most of yesterday's Meridian columns, including order value and delivery days.

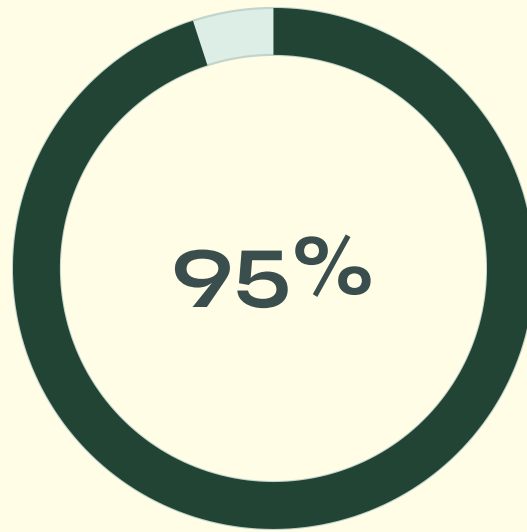
Normal: The Empirical Rule

This rule comes directly from the normal distribution. Meridian parcel weights: $\mu = 1.85$ kg, $\sigma = 0.42$ kg.



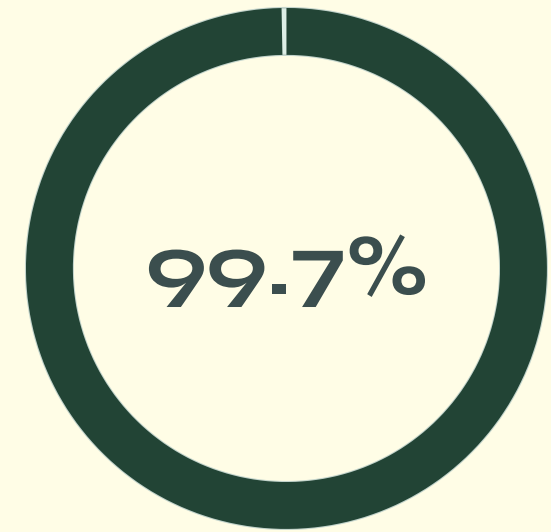
Within $\mu \pm 1\sigma$

Parcels between **1.43** and **2.27** kg



Within $\mu \pm 2\sigma$

Parcels between **1.01** and **2.69** kg



Within $\mu \pm 3\sigma$

Parcels between **0.59** and **3.11** kg

A parcel outside the 99.7% range is either a data entry error or something genuinely unusual – either way, it is worth investigating.

Normal: Z-Scores and Standardisation

The Formula

$$z = (\text{value} - \mu) / \sigma$$

The z-score converts any measurement into "how many standard deviations from the mean". This lets you compare across completely different variables and units.

Meridian Example

Parcel weights: $\mu = 1.85$ kg, $\sigma = 0.42$ kg. How unusual is a 2.5 kg parcel?

$$z = (2.5 - 1.85) / 0.42 = 1.55$$

A 2.5 kg parcel sits 1.55 standard deviations above the mean. From the normal table, the probability of exceeding that is **0.061** – about 6% of parcels are heavier than 2.5 kg.

 Z-scores only behave this way when the data is genuinely symmetric. On skewed data, they mislead. Always check the shape first.

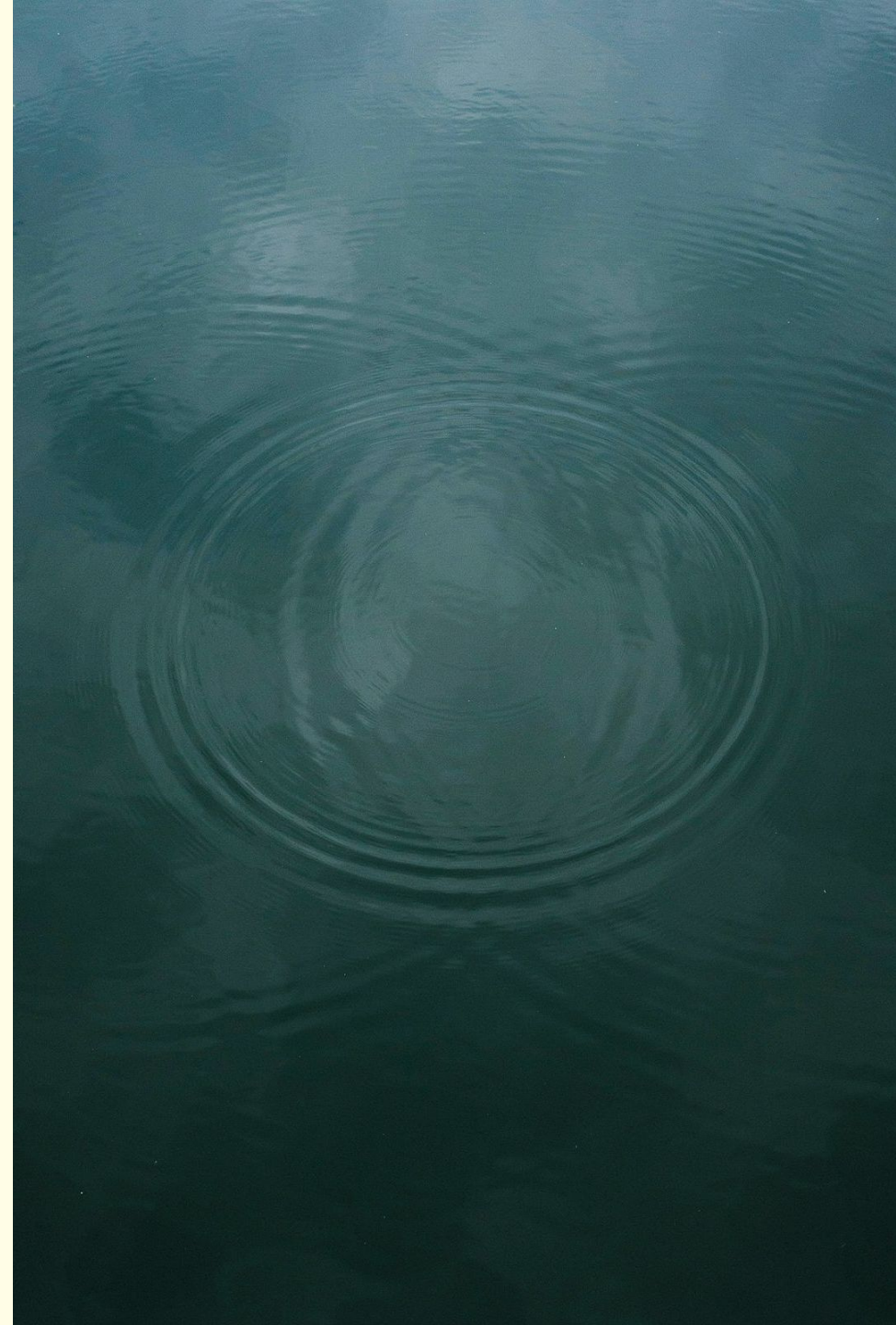
Normal: Why It Appears Everywhere

If you take repeated samples from almost any population and compute the average of each sample, those averages will follow a normal distribution – **even when the original data is not normal at all.**

This is the **Central Limit Theorem**. The sample needs to be reasonably large, usually taken as 30 or more.

- Average order value across 1,000 orders is approximately normal – even though individual order values are severely right-skewed
- Average daily sales, average defect rate, average response time – all approximately normal
- Almost every confidence interval and hypothesis test rests on this result

The distinction that matters: individual measurements are often not normal. Averages of them usually are.



Where Normal Is Applied



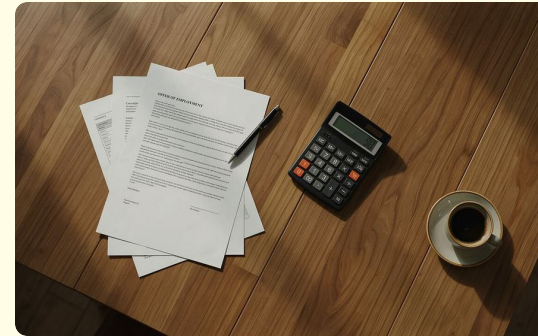
Manufacturing and Quality

Process control charts, tolerance limits, Six Sigma – all built on the normal distribution.



Finance

Portfolio risk models, Value at Risk, option pricing – with important caveats about tail behaviour.



HR and Education

Salary band design, standardised test scoring, and performance calibration frameworks.



Operations

Any physical measurement: weight, dimension, fill volume, temperature – where symmetry holds.

Normal: Advantages and Disadvantages

✓ Advantages

- Two parameters describe it completely – μ and σ
- The empirical rule gives fast, usable answers without software
- The Central Limit Theorem makes it apply to averages of almost anything
- Underpins nearly every standard statistical test – understanding it unlocks the rest of statistics

▪ Disadvantages

- **Symmetric only** – cannot model skewed data; yesterday's order value had skew of 12
- **Tails too thin for financial data** – real market returns produce far more extreme days than the model predicts
- **Allows negative values** – nonsense for weights, prices, and counts
- **The default people reach for** – applying it to skewed data is the most common mistake in applied statistics

PART 5

Putting It Together

How the four distributions connect – and how to choose between them.



How the Four Connect

They are not four unrelated tools. They form a chain – each one emerging from the previous under specific conditions.

1

Bernoulli

One trial, yes or no. The atomic building block.

2

Binomial

Add up n Bernoulli trials. A binomial with $n = 1$ is a Bernoulli.

3

Poisson

Take a binomial where n is very large and p is very small. This is why Poisson works for rare events across many opportunities.

4

Normal

Take a binomial or Poisson with a large enough average, and its shape becomes indistinguishable from a normal curve – the Central Limit Theorem appearing again.

Choosing the Right Distribution

Work down this table until you hit a match.

Your Question	Distribution
Will this one event happen, yes or no?	Bernoulli
Out of a fixed n trials, how many succeed?	Binomial
How many events in a fixed window, no fixed n ?	Poisson
What value will this measurement take?	Normal

Am I counting or measuring? Counting is never normal.	Do I have a fixed n? If no, it is not binomial.	Is my data symmetric? If no, it is not normal.
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Case Study – Meridian Retail, Day 2

Same company, same order book. Four questions, one for each distribution.

Known Facts from Day 1

- Site conversion rate: **28%** of visitors place an order
- Complaint calls to the support desk: **6 per hour** on average
- Parcel weight: mean **1.85 kg**, standard deviation **0.42 kg**, confirmed symmetric – skewness -0.05 , excess kurtosis -0.02

The Four Questions

1. What is the chance the **next single visitor** buys something?
2. Of the **next 20 visitors**, what is the chance that 8 or more buy?
3. What is the chance the support desk gets **10 or more calls** in an hour?
4. What fraction of parcels **exceed the 2.5 kg** standard shipping limit?

Case Study – Questions 1 and 2

Q1 – Next Single Visitor → BERNOULLI

One trial, two outcomes. $p = 0.28$.

- $P(\text{buys}) = \mathbf{0.28}$
- $P(\text{does not buy}) = \mathbf{0.72}$

Q2 – Next 20 Visitors, 8 or More Buy → BINOMIAL

Fixed $n = 20$, constant $p = 0.28$, independent visitors.

- Expected conversions = **5.6**, SD = **2.01**
- **$P(8 \text{ or more}) = 0.171$**
- $P(\text{exactly } 5) = 0.193 \mid P(2 \text{ or fewer}) = 0.053$

📌 An 8-conversion day happens about 17% of the time – roughly once a week. It is a good day, not evidence that a campaign worked. Distinguishing normal variation from a real effect is exactly what A/B testing does.

Case Study – Questions 3 and 4

Q3 – 10 or More Calls in an Hour → POISSON

Calls arrive at a rate; there is no fixed number of trials. $\lambda = 6$.

- $P(\text{exactly 6 calls}) = 0.161$
- **$P(10 \text{ or more}) = 0.084$**
- $P(12 \text{ or more}) = 0.020$

Staffing for the average of 6 leaves the desk overwhelmed roughly 8% of hours. Staffing for **10 covers 95%** of hours. Plan against the tail, not the average.

Q4 – Parcels Over 2.5 kg → NORMAL

A measurement, and the column is symmetric – so normal applies.

- $z = (2.5 - 1.85) / 0.42 = \mathbf{1.55}$
- **$P(\text{over 2.5 kg}) = 0.061$**

About **6% of parcels** breach the standard rate and attract a surcharge – a directly actionable number for pricing decisions.

Case Study – Checking the Model Against Reality

A model is only worth using if it matches what actually happened. We have 5,000 real parcels, so we can check directly.

Check	Normal Model Predicts	Actual Data
Parcels over 2.5 kg	6.28%	6.08%
Parcels between 1.4 and 2.3 kg	71.08%	71.44%

The model is accurate to within a fraction of a percent. **Why it worked:** parcel weight is genuinely symmetric – skewness -0.05 , excess kurtosis -0.02 , verified yesterday.

Where it would have failed: applying the same normal model to `order_value` (skewness 12.2) or `delivery_days` (skewness 8.3) would produce confident, precise, badly wrong answers.

 Always check the shape before choosing the distribution. That check is yesterday's material, and it is what makes today's material safe to use.

Common Mistakes

Normal for Count Data

Number of calls cannot be 6.3, and cannot be negative. Counts are never normal.

Binomial Without Fixed n

"How many complaints tomorrow" has no n . That is a Poisson question – not binomial.

Poisson When Overdispersed

If variance far exceeds the mean, Poisson understates the tail. Check the two numbers first.

Assuming Independence

Group bookings, viral products, and queues all break independence – silently.

Normal on Skewed Data

The most common and most expensive error. Familiarity is not justification.

Reporting Without Stating Assumptions

"There is a 17% chance" means nothing until you state what you assumed to get there.

Cheat Sheet – The Four Distributions

	Bernoulli	Binomial	Poisson	Normal
Type	Discrete	Discrete	Discrete	Continuous
Inputs	p	n, p	λ	μ, σ
Mean	p	np	λ	μ
Variance	$p(1-p)$	$np(1-p)$	λ	σ^2
Use for	One yes/no	Successes in n trials	Events per window	Measurements

```
from scipy import stats
stats.bernoulli.pmf(1, p=0.28)    # P(success) on one trial
stats.binom.pmf(5, n=20, p=0.28)  # P(exactly 5 of 20)
1 - stats.binom.cdf(7, n=20, p=0.28) # P(8 or more)
stats.poisson.pmf(6, mu=6)        # P(exactly 6 calls)
1 - stats.poisson.cdf(9, mu=6)    # P(10 or more)
stats.norm.cdf(2.5, loc=1.85, scale=0.42) # P(weight below 2.5)
```

📌 **The one line to take away:** check the shape of your data before you choose a distribution – because every answer a distribution gives you inherits the assumption you made when you picked it.